

LMM NN-Surrogate — Interview Cheat Sheet

One-line pitch

Trained a neural network to replace the Monte-Carlo LMM pricer **inside the calibration loop** for the Callable Exotics book (CRAs, CDRANs, Bermudans). Used by Valuation Control for IPV. End-to-end calibration: **overnight** → **under an hour**, ~100×, with every result re-checked by the real MC pricer.

What the NN actually learns

Block	Fields (what's in a training row)
A. LMM params <i>(optimiser varies these)</i>	σ -curve (a, b, c, d); SABR (α , β , ρ , ν) per tenor bucket; correlation (ρ_∞ , β_{corr}); shift
B. Instrument descriptor	type flag (caplet / swaption), expiry T, tenor, strike K, $\log(K/F)$, payer/receiver
C. Market context <i>(frozen per snapshot)</i>	F at expiry, F_atm, discount factor, curve level + slope
Target	Black-implied vol (not price) from full LMM MC pricer

Key scope point: the NN prices only **caplets and swaptions** (the calibration instruments) — not the CRAs/CDRANs/Bermudans themselves. Those are still priced by full MC using the calibrated params.

Example row

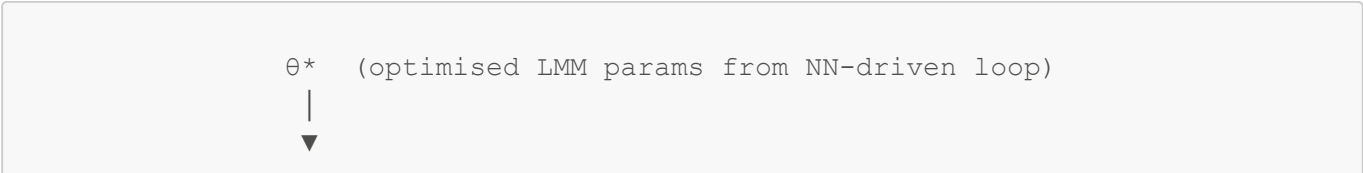
type	T	tenor	K	$\log(K/F)$	σ_a	sabr_α	ρ_∞	target_iv
swaption	5	5	3.50%	-0.04	0.18	0.015	0.30	0.2814

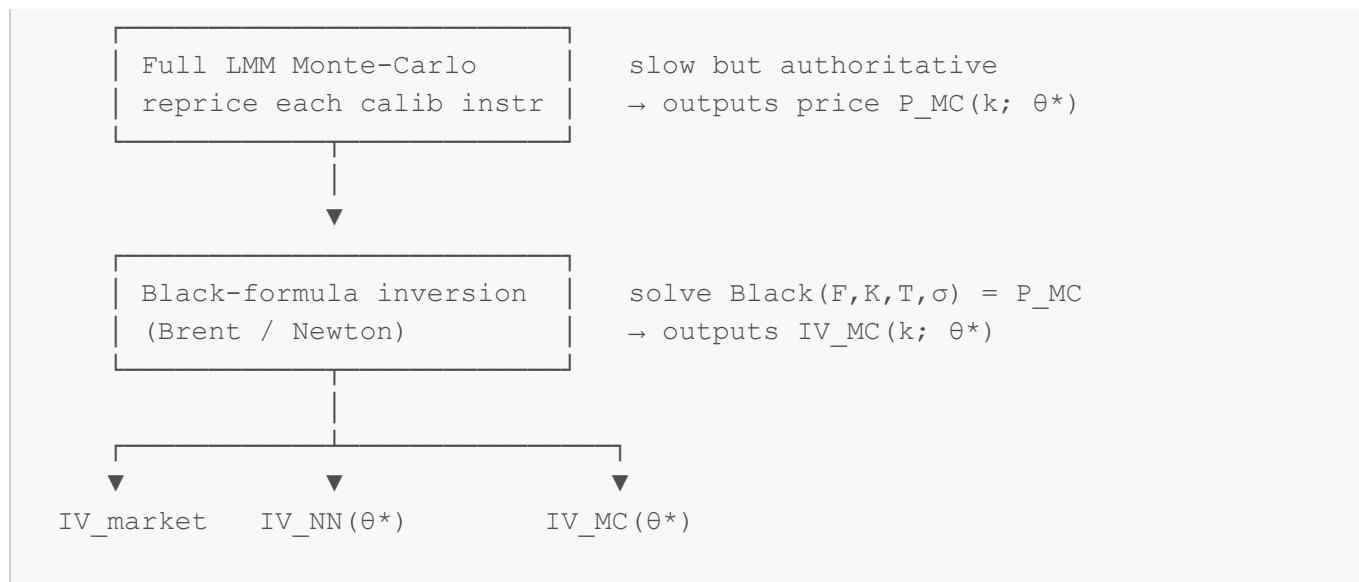
Same row with K=4.00% → IV=0.2997 (NN learns the **smile**). Same row but caplet, T=1, K=3.00% → IV=0.2206 (NN learns the **product type**).

Why Black IV (not price)

- Bounded, smooth, dimensionless → trains far better
- Same units as market quotes and verification output → like-for-like residuals
- Smile diagnostics fall out for free when plotted against log-moneyness

Verification — how the calibration is checked





The Black inversion in one sentence

Take the MC's output price, hold $(F, K, T, \text{annuity})$ fixed, solve for the σ that reproduces it under Black's formula — that σ is the MC's implied vol, directly comparable to market and to the NN's prediction.

Two residuals

Residual	Question	Tolerance
$IV_{market} - IV_{MC}(\theta^*)$	Is the LMM well-fitted to market?	< 25–50 bp per instrument, RMSE < 15 bp
$IV_{NN}(\theta^*) - IV_{MC}(\theta^*)$	Was the NN reliable at θ^* ?	< 5–10 bp

First test = Valuation Control acceptance. Second test = model-risk acceptance for the surrogate.

Worked example

5y × 5y payer swaption, $K = 3.50\%$, market IV = **28.20 %**:

Stage	Output
NN inside optimiser	$IV_{NN} = \mathbf{28.14\%}$
Full MC at θ^*	$P_{MC} = 0.02476$ (price)
Black inversion of P_{MC}	$IV_{MC} = \mathbf{28.17\%}$
Calibration residual (market – MC)	+3 bp ✓
Surrogate residual (NN – MC)	–3 bp ✓

Both pass → calibration accepted, θ^* handed to Valuation Control. If surrogate residual had been +60 bp → reject: NN mispriced near θ^* → retrain or fall back to MC-in-loop.

Timings & speed-up

Step	Base (MC in loop)	NN surrogate
Inner pricing call	30 s – 3 min × 60 instruments × ~200 iter	~5 ms total per iteration
Optimiser wall time	8–24 hours	~1 second
Verification sweep (full MC at θ^*)	—	~30 min
End-to-end	8–24 hours	~30–45 min

Headline: ~100× end-to-end. Inner loop alone is orders of magnitude faster — but verification is the binding constraint and model risk requires it.

Interview soundbites (recite these)

1. *"The NN doesn't price the trades — it prices the calibration instruments. The trade book is still valued by full MC using the calibrated parameters."*
2. *"Target is Black implied vol, not price — bounded, smooth, and in the same units as market quotes and the verification output."*
3. *"After the optimiser converges, we plug θ^* back into the full MC pricer, get a price, invert Black to recover an MC-implied vol, then compare three things: market, NN, MC. Two residuals fall out — one tests the model fit, one tests the surrogate."*
4. *"100× end-to-end. The optimiser loop alone is much faster than that, but the MC verification sweep at the end is the floor — and removing it was never on the table for Valuation Control."*
5. *"Same architecture covered CRAs, CDRANs and Bermudans — they share the LMM, so they share the calibration, so they share the surrogate."*

CV bullet (≤30 words)

LMM Calibration Surrogate (NN): Replaced Monte-Carlo pricing inside the LMM calibration loop for CRAs/CDRANs/Bermudans — ~100× faster IPV calibration for Valuation Control's Callable Exotics book, accuracy benchmarked against full LMM.